Subfield planes and unordered pairs of parabolics

Human Compression Project

Abstract

We count two families from the fixed Part II classification of rank two geometries for $\mathrm{PSL}(2,q)$. In characteristic two, planes through a fixed subfield line give the parameter space; Frobenius gives the equivalence relation. This recovers the numerical formulas for the $A_5, E_{16} : 3$ family and extends its parameter description to the previously omitted subfield covers. In characteristic three, unordered pairs in an additive quotient give the family with two A_4 parabolics. This halves its printed conjugacy count under the source's stated convention, while preserving its automorphism count. Classical subgroup and automorphism theorems remain explicit premises.

1 Local actions and the counting convention

Let G be a finite simple projective group. For distinct proper subgroups H, Q, put $B = H \cap Q$. The rank two coset geometry has the cosets of H, Q as its two types, with nonempty intersection defining incidence. The two local actions are those of H, Q on their cosets of B. When both are two-transitive and one parabolic is maximal in G, these subgroups generate G and give a connected, flag-transitive, residually weakly primitive geometry. We use the proof of this assertion in the previously accepted projective-line manuscript; it depends only on coset actions and maximality.

The fixed Part II source, Section 2 on physical page 4, defines conjugacy by equality of the *sets* $\{H^g, Q^g\}$ and $\{H', Q'\}$. Thus interchanging the two types is allowed. We call this the unordered convention. We will give ordered counts separately when they differ. Equivalence under $\mathrm{Aut}(G)$ uses the same convention.

We retain the following classical facts used by the source: the natural subfield embeddings and their maximality at prime extension degree; the conjugacy of those embeddings in the cases specified below; $\mathrm{Aut}(\mathrm{PSL}(2,p^f)) = \mathrm{P\Gamma L}(2,p^f)$ for the ambient groups used here; and the usual projective orders. These are not assumed consequences of the geometry classification under audit. Every normalizer needed for the new counts is proved below.

2 Planes through a subfield line

Let $K = \mathbb{F}_s \subset F = \mathbb{F}_{s^m}$, with $s \geq 4$ even and m prime. Write $U_T = \{x \mapsto x + t : t \in T\}$ and $C = \{x \mapsto ax : a \in K^*\}$. Set

$$G = \mathrm{PSL}(2,F) = \mathrm{PGL}(2,F), \quad H = \mathrm{PSL}(2,K), \quad B = U_K C.$$

For a two-dimensional K-subspace T containing K, put $Q_T = U_T C$. The earlier accepted subfield-plane proof establishes

$$H \cap Q_T = B, \qquad |H| = s(s^2-1), \quad |Q_T| = s^2(s-1), \quad |B| = s(s-1),$$

and proves all geometry hypotheses. Its argument is a scalar-subspace correspondence: groups between $U_K C$ and $U_T C$ are the groups $U_W C$ for K-subspaces $K \subseteq W \subseteq T$. A cover has quotient dimension one and induces $\mathrm{AGL}(1, K)$ on its local residue. No subfield of degree two is needed.

Lemma 1. *The normalizer of B in G is B. For each conjugate of B, there is exactly one conjugate of H containing it. The conjugacy classes of the displayed geometries are consequently indexed by $\mathrm{PG}(m-2, s)$, the lines of F/K.*

Proof. The translation subgroup U_K is the unique Sylow two-subgroup of B, and its common projective fixed point is infinity. A normalizer of B is therefore affine, say $x \mapsto ax+b$. Normalizing U_K requires $aK = K$, hence $a \in K^*$. Conjugating any nonidentity scalar in C then requires $(1-c)b \in K$, so $b \in K$. This proves the normalizer assertion, including its converse.

The subfield groups H form one G-class in this even-characteristic case. Their point stabilizers form one class inside H. If H^g contains B, conjugate a point stabilizer inside H before applying g; the resulting transporter normalizes B. It belongs to B by the first paragraph, so $H^g = H$.

To compare two displayed pairs, the unequal parabolic orders first force H to be preserved. Their intersections then force B to be preserved. The transporter is in $N_G(B) = B$. Its translations lie in $K \subset T$ and its scalars in K^*, so it normalizes each Q_T. Hence two planes give conjugate pairs exactly when they are equal. The planes containing K correspond to the lines of F/K, as asserted. □

Proposition 2. *There are exactly $(s^{m-1}-1)/(s-1)$ conjugacy classes in this family. Their $\mathrm{Aut}(G)$-classes are the orbits of the absolute Frobenius $x \mapsto x^2$ on $\mathrm{PG}(m-2, s)$.*

Proof. The conjugacy count is Lemma 1. Every field automorphism preserves the standard H and B. Write an arbitrary ambient automorphism as an inner automorphism followed by a field automorphism. If it transports one of our pairs to another, its inner part must normalize B and therefore belongs to B. This part acts trivially on the planes. Conversely every Frobenius power gives the stated transport. This proves the orbit description, with no assertion that the field automorphism action is free. □

Lemma 3. *An affine subgroup Q containing B with B maximal in Q is one of the groups Q_T above. In particular every such affine extension has a two-transitive local action.*

Proof. Write $V = Q \cap U_F$ for its translations and D for its image in F^*. Both V and K are stable under K^*, so V is a K-space. The subgroup $U_V C$ lies between B and Q. If $V > K$, maximality forces $Q = U_V C$, and the scalar correspondence forces $\dim_K(V/K) = 1$. If $V = K$, an element $x \mapsto ax + b$ of Q normalizes U_K, so $a \in K^*$. Multiplying by its scalar inverse in C leaves translation by $b \in V = K$. Thus $Q = B$, contrary to the proper extension. This excludes the second case. □

3 The $A_5, E_{16} : 3$ formulas

Specialize to $K = \mathbb{F}_4$, $F = \mathbb{F}_{4^r}$ with r prime. The numbers in this section are the ones asserted by Lemmas 5.3–5.4 and the corresponding rows of Table 1 in the fixed source.

Lemma 4. *For odd prime r, the groups isomorphic to $E_{16} : 3$ in G form $(4^{r-1}-1)/15$ conjugacy classes.*

Proof. Such a group has characteristic normal subgroup E_{16}, whose nonidentity elements are unipotent and have a unique common fixed projective point. Put this point at infinity. Its complement of order three has a second fixed point; a translation puts that point at zero. The group is now $U_T C$ for a K-plane $T \subset F$. Conversely each such plane supplies the stated group. Affine conjugation identifies exactly the scalar multiples of the planes: translations can change a chosen complement but cannot change the translation subgroup.

The multiplier ring $E_T = \{a \in F : aT \subseteq T\}$ is a field. Closure under addition and multiplication is immediate; for nonzero $a \in E_T$, multiplication is a bijection of the finite set T, so $a^{-1} \in E_T$. It contains K, and $[E_T : K]$ divides both $\dim_K T = 2$ and $[F : K] = r$. Thus $E_T = K$. Every scalar orbit has length $(4^r - 1)/3$. There are

$$\begin{bmatrix} r \\ 2 \end{bmatrix}_4 = \frac{(4^r - 1)(4^r - 4)}{(16 - 1)(16 - 4)}$$

planes, by counting ordered independent pairs and dividing by the number of ordered bases of a plane. Division by the orbit length gives $(4^{r-1} - 1)/15$. □

Theorem 5. *For $r = 2$, the family with parabolics $A_5, E_{16} : 3$ has one class under each equivalence. For $r = 3$ it has five conjugacy classes and two $\mathrm{Aut}(G)$-classes. For prime $r > 3$ the respective counts are*

$$N = \frac{4^{r-1} - 1}{3}, \qquad \frac{N + (2^{r-1} - 1)}{2r} = \frac{2(4^{r-2} - 1) + 3 \cdot 2^{r-2}}{3r}.$$

Proof. The planes through K form $\mathrm{PG}(r - 2, 4)$, of size N; when $r = 2$ there is just the plane F. Put $V = F/K$, and let σ be squaring on this quotient. We must count its orbits, by Proposition 2. It has order dividing $2r$. Put $\tau = \sigma^2$, the K-linear Frobenius of order r, and $\beta = \sigma^r$, an involution semilinear over $K/\mathbb{F}_2$.

First suppose $r > 3$. If a line is fixed by τ, choose a nonzero vector v on it with $\tau v = av$, $a \in K^*$. Iterating gives $a^r = 1$, so $a = 1$ because r is prime and different from three. Writing $v = x + K$ gives $x^4 - x = c \in K$. The relative trace from F to K annihilates the left side and takes c to $rc = c$, so $c = 0$. Then $x \in K$, a contradiction. Every nonidentity power of τ has the same fixed lines. An odd power of σ other than β generates the cyclic group of order $2r$, so it has no fixed line either.

It remains to count the lines fixed by β. If $\beta(v) = av$, then $a \in K^*$ and one may replace v by $a^{-1}v$ to make it fixed: $\beta(a^{-1})a = a^{-1}$ since $a^3 = 1$. Each stable K-line therefore contains exactly one nonzero fixed vector. The fixed quotient space V^β is naturally $\mathbb{F}_{2^r}/\mathbb{F}_2$. Indeed, for a fixed class $x + K$, $c = \beta x - x \in K$ satisfies $\beta c = c$, hence $c \in \mathbb{F}_2$. The map $k \mapsto k^2 - k$ from K onto $\mathbb{F}_2$ provides k with $\beta k - k = c$; subtracting k gives a representative fixed by β. Its ambiguity is exactly $\mathbb{F}_2$. Thus β fixes $2^{r-1} - 1$ projective points. Burnside's formula gives the claimed average over $2r$ powers.

For $r = 3$, V has dimension two over K. The trace argument still shows that τ has no fixed nonzero vector. Since $\tau^3 = 1$, it is semisimple and its possible eigenvalues are the two nonidentity cube roots in K. The commuting semilinear map β squares eigenvalues and is invertible, so both eigenspaces occur and have the same dimension. Each therefore has dimension one. These are precisely the two invariant lines, and β interchanges them. The odd generators σ, σ^5 also interchange them and fix no line. The preceding fixed-quotient argument gives three lines fixed by β. The six fixed-point counts are $5, 0, 2, 3, 2, 0$, whose average is two. □

4 Two A_4 parabolics in characteristic three

Let $q = 3^m$, where m is an odd prime, and put $G = \mathrm{PSL}(2, q)$, $H = \mathrm{PSL}(2, 3) = A_4$, $B = U_{\mathbb{F}_3} \cong C_3$. The classical subfield theorem makes H maximal in G; its conjugates form one class. Put $V = \mathbb{F}_q/\mathbb{F}_3$, regarded as an additive group.

Lemma 6. *The conjugates of H containing B form a regular V-set under translations. Distinct such subgroups intersect exactly in B. Conjugacy classes of ordered pairs of them are indexed by $V \setminus 0$; unordered classes are indexed by $(V \setminus 0)/\{v \sim -v\}$.*

Proof. Any normalizer of B fixes its unique projective fixed point and is affine. Its slope must lie in $\mathbb{F}_3^* = \{1, -1\}$. An affine transformation belongs to $\mathrm{PSL}(2, q)$ precisely when its slope is a square in $\mathbb{F}_q$. Since m is odd, $q \equiv 3 \pmod 4$ and -1 is not a square. Therefore $N_G(B) = U_{\mathbb{F}_q}$.

Every copy of B inside H is conjugate under H, and $N_H(B) = B$. Thus the copies of H containing the fixed B are exactly its translates $H_t = u_t H u_t^{-1}$, with equality precisely when $t - t' \in \mathbb{F}_3$. Here we also use $N_G(H) = H$, which follows from maximality and simplicity. Translation acts regularly on the index set V. Since B is maximal in A_4, any two distinct H_t meet in B.

Every eligible pair of A_4 subgroups with intersection of order three can be transported to this set, since its intersection is a conjugate of B. Any transporter between two such pairs must normalize their intersection. Hence it acts as translation on V. Translating an ordered pair (a, b) to $(0, b-a)$ gives precisely its nonzero difference. Interchanging the entries changes this difference to its negative. These are all identifications, proving both parameterizations. □

Theorem 7. *Under the unordered convention of the fixed source, the number of conjugacy classes of these geometries is*

$$\frac{3^{m-1} - 1}{2}.$$

The number of $\mathrm{Aut}(G)$-classes is two if $m = 3$, and $(3^{m-1} - 1)/(2m)$ otherwise. The ordered conjugacy count is $3^{m-1} - 1$.

Proof. Distinct pairs give the required geometries: their intersection is C_3, each local A_4 action on four cosets is two-transitive, and maximality of either A_4 gives connectedness and the residual primitivity conditions. In characteristic three a nonzero vector is different from its negative. Lemma 6 therefore proves the two conjugacy counts.

In $\mathrm{P\Gamma L}(2, q)$ the normalizer of B consists exactly of affine maps with slope ± 1, followed by field automorphisms. The standard H is preserved by sign and by every field automorphism. It follows that the additional action on differences is generated by $v \mapsto -v$ and $\phi : v \mapsto v^3$. Its group is $C_2 \times C_m$. Inversion from interchanging the two parabolics is already in this group.

Suppose first that $m \neq 3$. If $\phi(x + \mathbb{F}_3) = x + \mathbb{F}_3$, then $x^3 - x = c \in \mathbb{F}_3$. Relative trace gives $mc = 0$, whence $c = 0$ and $x \in \mathbb{F}_3$. Thus no nonzero vector is fixed by ϕ, or by any of its nonidentity powers. A map $-\phi^k$ fixes no nonzero vector either: its mth power is negation. Consequently the action on $V \setminus 0$ is free, proving the formula.

For $m = 3$, V has dimension two over $\mathbb{F}_3$. The induced ϕ has order dividing three and is not the identity. To see the latter, if every $x \in \mathbb{F}_{27}$ satisfied $x^3 - x \in \mathbb{F}_3$, the additive map $x \mapsto x^3 - x$ would have image of size at most three, whereas its kernel has size three and its image has size nine. Thus ϕ has one nontrivial unipotent Jordan block on V, and its fixed subspace has dimension one. The two nonidentity powers of ϕ each fix two nonzero vectors. The three maps $-\phi^k$ fix none. Burnside's formula gives $(8 + 2 + 2)/6 = 2$. □

5 Source joins and attribution

Theorem 5 recovers the $A_5, E_{16} : 3$ counts printed in Table 1, rows 2–4, and Lemmas 5.3–5.4 of the published Part II paper. Lemma 4 separately counts subgroup classes, not geometries; these two counts must not be confused. The same mechanism supplies the parameter space for the omitted higher-extension subfield covers already proved in the accepted correction. It does not repeat that acquisition.

Theorem 7 corrects the inner conjugacy column in Table 5: the printed $3^{m-1} - 1$ becomes $(3^{m-1} - 1)/2$, and the printed eight at $q = 27$ becomes four. Both printed automorphism counts remain correct. The separately fixed 21-page appendix repeats the ordered count in its proof of Proposition 26, physical page 15. Its calculation counts the possible second parabolics after choosing a first one; it omits the additional identification required by the source's unordered definition. The explicit conjugator u_{-t} carries the unordered pair $\{H_0, H_t\}$ to $\{H_{-t}, H_0\}$. This gives the missing identification inside G, without a computational assumption.

The original numerical formulas recovered here belong to Buekenhout, De Saedeleer and Leemans. Gaussian counts, finite-field trace, Burnside's formula and normalizer actions are standard. The contribution of this note is the common projective parameter description, its extension to the already established missing covers, and the explicit audit of the unordered convention. No historical priority beyond this fixed-source comparison, complete treatment of either original, Lean verification, new open-conjecture answer or compression measurement is claimed.

References

[1] F. Buekenhout, J. De Saedeleer and D. Leemans, *On the rank two geometries of the groups* PSL$(2, q)$*: part II*, Ars Mathematica Contemporanea **6** (2013), 365–388.

[2] The same authors, *Part II (APPENDIX)*, the fixed separately catalogued 21-page edition, proof of Proposition 26.

[3] Human Compression Project, *Subfield planes in projective rank two geometries*; and *Local actions in rank two over a projective line*, previously accepted components.